

SECTION B

6 (short)

- (a) A combinational logic design has four inputs M, N, O, P. The single output Y is HIGH if two or more inputs are HIGH. Otherwise Y is LOW. Using a Karnaugh map, derive the expressions for Y in Sum of Products (SoP) and Product of Sums (PoS) form. [6]
- (b) Modify the expressions in (a) using De Morgan's theorem for NAND and NOR gate only implementations. Assuming gates with any number of inputs are available, which implementation will require fewer gates? [4]

7 (short)

- (a) Figure 5 shows a 4-bit Digital to Analog converter (DAC).
- (i) Identify the Least Significant Bit (LSB) and Most Significant Bit (MSB) inputs. [1]
- (ii) If the op-amp is ideal and $R = 1 \text{ k}\Omega$, what are the input impedances at N1 and P1? [2]
- (iii) Calculate V_{out} when all the inputs are connected to HIGH (5 V). [2]
- (b) What are the drawbacks of this design? With a brief explanation, suggest an alternative DAC architecture that does not have these drawbacks. [5]

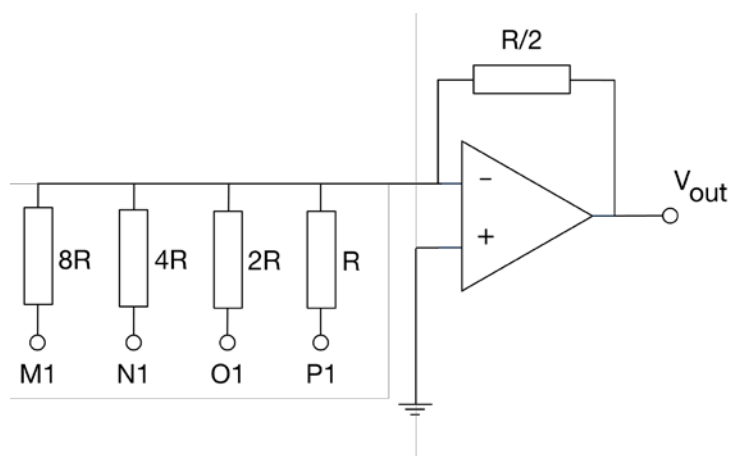


Fig. 5